

Markov State Modeling of the HP35 Domain

Final project for the course

Biological Datasets for Computational Physics

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I. INTRODUCTION

The villin headpiece subdomain HP35 is one of the fastest folding proteins. The Lys24-Nle/Lys29-Nle double mutant studied here was experimentally found to fold in $\simeq 0.7 \mu\text{s}$ at 300 K with little temperature dependence [1]. Its native fold is a compact three-helix bundle whose stability derives largely from a hydrophobic core formed by the aromatic residues *Phe6*, *Phe10* and *Phe17* (Fig.1). Its short folding time made it possible, already in the early 2010s, to run long unbiased all-atom simulations resolving dozens of folding and unfolding events [2], establishing it as a paradigmatic model system for protein folding. Both experimental [3] and simulation based studies [4–6] have reported evidence of structurally diverse long-lived states within its conformational ensemble.

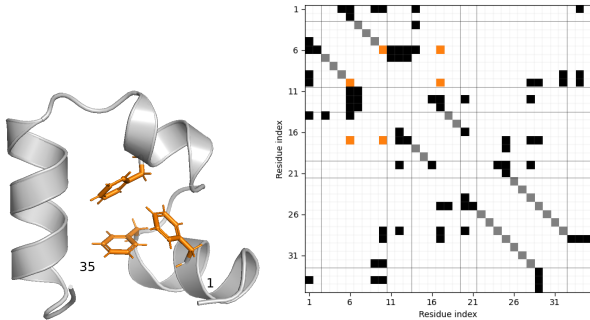


FIG. 1: Crystal structure and contact map of HP35 (PDB: 2F4K [1]). Left: The folded structure consists of three α -helices (3-10), (14-19), and (22-32), connected by two loops. Highlighted in orange are aromatic residues *Phe6*, *Phe10*, *Phe17*, whose interactions contribute to the formation of the hydrophobic core. Visualization with PyMOL [18]. Right: contact map computed with a distance cutoff of 4.5 Å. Highlighted in grey are the intra-helix contacts ($i, i + 4$), and in orange the contacts between the three mentioned aromatic residues.

In this project, I analyse the well-known 305 μs trajectory of the Lys24-Nle/Lys29-Nle mutant published by Lindorff-Larsen *et al.* in 2011 [2]. On this trajectory I build Markov State Models (MSMs) [7] to ask whether

HP35 possesses resolvable metastable substructure.

The MSM pipeline (feature selection, dimensionality reduction, clustering) has no unique optimal recipe (yet) [6, 8]. As a consequence, different seemingly reasonable choices may resolve different metastable states and populations. This motivates a deliberately redundant design: rather than commit to a single pipeline, I build four MSMs from two alternative internal-coordinate representations (residue-residue contacts and backbone dihedral angles), each reduced by two alternative methods (PCA and tICA). I will then use spectral analysis and PCCA++ coarse-graining to identify long-lived states and characterize them structurally.

II. METHODS

A. Dataset assembly

I use the preprocessed HP35 trajectory distributed by Nagel *et al.* [8, 9], derived from the original simulations of Lindorff-Larsen *et al.* [2]. The dataset provides several internal-coordinate representations specifically selected and filtered for MSM construction (see [9] for feature definitions). Each time series contains $\simeq 1.5 \times 10^6$ frames at a sampling interval of $\simeq 200$ ps.

I employ two of these representations as alternative feature sets: residue-residue contact distances (`hp35.mindists2`, 42 features) and backbone dihedral angles (`hp35.dihs.shifted`, 62 features).

A third set of crystal-structure contact distances (`hp35.crystaldists`, 53 features) is used as an additional structural descriptor in later analysis (e.g. to compute an approximate RMSD from the crystal structure). To enable structural visualization in absence of cartesian coordinates, I reconstructed the backbone atom coordinates from the raw dihedral-angle time series (`hp35.dihs`) using MDTraj [19]. All molecular renderings shown below are obtained this way.

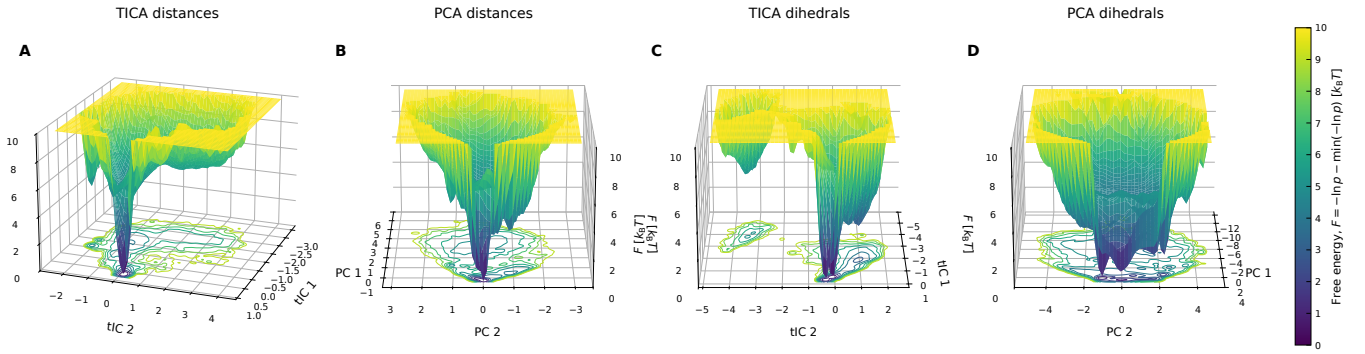


FIG. 2: *Two-dimensional free-energy surfaces obtained by projecting the data onto the first two reduced coordinates. Distance-based models exhibit a single dominant minimum, whereas dihedrals-based models show two major minima. The tICA projections additionally contain a low-free-energy basin well separated from the main funnel. Because free energy surfaces do not have a clear interpretation in terms of dynamics, these landscapes should be interpreted qualitatively. Nevertheless, substantial differences are observed among the four pipelines which remain somewhat still observable in the metastable structure resolved by the MSMs (Sec. III D, III E).*

B. Methods for MSM construction

All code is written in Python, using Deeptime [11] for MSM construction and validation and BioPython [12] for analysis of the crystal structure (PDB 2F4K [1]).

A Markov state model discretizes the continuous high-dimensional dynamics into a finite set of states with transitions observed at a fixed time interval (τ_{MSM}), aiming for a reduced representation that preserves the slow dynamical processes of the original system. The pipeline used here consists of the following steps:

1. Feature selection (distances or dihedrals),
2. Dimensionality reduction (PCA or tICA),
3. Clustering of the reduced coordinates into discrete states (k-means),
4. Estimation of the transition matrix and selection of the Markov lag time τ_{MSM} (implied-timescale convergence test; see [7, Ch. 2.4.1]),
5. Validation against the data (Chapman–Kolmogorov test; see [7, Ch. 4.8]).

The first two steps are the ones explored most in this work. Together they compress the raw simulation data into a low-dimensional space where clustering becomes feasible. Since k-means relies on Euclidean distance, rapidly interconverting conformations must map to nearby points in the reduced space; failure to satisfy this condition typically leads to poor markovianity [13].

Feature selection. Two feature sets are used: contact distances (`hp35.mindists2`) and backbone dihedrals (`hp35.dihs.shifted`). Following [9], a Gaussian low-pass filter ($\sigma = 2$ ns) is applied to each feature to suppress high-frequency recrossings between states.

Dimensionality reduction. Two linear methods are compared: PCA, which finds directions of maximal variance,

and tICA, which finds directions of maximal autocorrelation at a lag time τ_{tICA} . PCA therefore captures the largest structural fluctuations, whereas tICA emphasizes the slowest dynamical processes: configurations far apart in tICA space tend to interconvert slowly, giving a closer correspondence between geometric and kinetic separation (see Supp. Methods BS1). Because MSMs aim to resolve slow kinetics, tICA is generally preferred. Combining the two feature sets with the two reduction methods gives four pipelines: *tica-distances*, *tica-dihedrals*, *pca-distances*, and *pca-dihedrals*. Since HP35 folds on the microsecond timescale [1, 2], τ_{tICA} was inspected over 100–1000 frames ($\simeq 20$ –200 ns), well below the folding time, and fixed at 20 ns after checking that the leading tICA components were stable across this range. The number of retained components was selected using a heuristic criterion aimed at identifying reduced coordinates capable of separating metastable states. Relevant components are expected to exhibit multimodal free-energy profiles [9] and time series characterized by long-lived plateaus separated by relatively rapid transitions [14]. Applying this criterion led to the retention of 7 components for the *tica-dihedrals* model and 5 components for the remaining three pipelines (Supp. Fig. SF1).

Clustering and lag-time selection. The reduced coordinates were clustered with k-means (k-means++ initialization, up to 100 iterations), testing $k = 30, 50$, and 100. The number of clusters and the lag time were chosen together, since they jointly control resolution and statistical reliability: more clusters provide higher resolution and speed up implied-timescale convergence, but leave fewer transitions per microstate and can disconnect the microstate network. The lag time must also stay well below the microsecond folding time [2], or intermediate kinetic details are lost. Because the implied-timescale test alone does not fix the lag time, the final choice was a com-

promise between timescale convergence, network connectivity, and adequate sampling of transitions, and so remains partly qualitative. Transition matrices were then estimated by maximum likelihood with a sliding-window scheme [7, Ch. 4.1] enforcing detailed balance, retaining the largest connected component as the state space. From a grid search with $n_{clusters} = \{30, 50, 100\}$ clusters and lag times between 10 and 100 ns; $n_{clusters} = 50$ and a lag time of 50 ns is selected for all four pipelines, as this seemed the best compromise between resolution, markovianity and connectivity (Supp. Fig. SF2 and SF3).

Spectral analysis and coarse-graining. Perron cluster-cluster analysis (PCCA) exploits the dominant eigenvectors of a Markov transition matrix to identify metastable sets of states [7, Ch. 2.5.1–2.5.2]. The number of macrostates is chosen from the implied-timescale spectrum, taking the largest gap between successive timescales as an estimate of the number of dominant slow processes. Microstates are then lumped into macrostates according to their contributions to the slow relaxation modes of the system. The resulting macrostates are interpreted as metastable regions of conformational space, between which probability exchange occurs on slow timescales [7, Ch. 2.5.1–2.5.2]. In this work I use PCCA++ (robust Perron cluster-cluster analysis), the variant implemented in Deeptime [11]. PCCA++ assigns to each microstate i a vector of fuzzy memberships $\chi_{ik} \in [0, 1]$, which quantify the degree to which the microstate belongs to each macrostate k , with $\sum_k \chi_{ik} = 1$. Each trajectory frame inherits the membership vector of the microstate to which it is assigned.

Dynamical coring. Assigning each frame to its highest-membership macrostate produces a hard state trajectory, but near a state boundary rapid recrossings inflate the apparent number of transitions: these are more likely to reflect boundary noise than genuine interstate crossings. Dynamical coring [16] is a heuristic procedure to suppress them: starting from the hard-assigned trajectory, a transition into a new macrostate is registered only once the trajectory remains there for at least a threshold time $t_{\min}$, so that only long-lived occupancies survive. In the following I applied coring to the discretized macrostate trajectories with $t_{\min}$ equal to one lag time ($t_{\min} = \tau_{\text{MSM}} = 50$ ns).

C. Statistical analysis

Chapman–Kolmogorov test. Model self-consistency was assessed with the CK test as implemented in Deeptime [11]. The microstate MSM at $\tau_{\text{MSM}} = 50$ ns was coarse-grained into four metastable sets with PCCA++; for a range of lag times $k\tau_{\text{MSM}}$, the coarse-grained transition probabilities predicted by the reference model (propagated to $k\tau_{\text{MSM}}$) were compared with those estimated

independently at each lag. Predicted and independent curves track closely and relax to consistent stationary values (Supp. Fig. SF4, shown for *pca-dihedrals*; the others behave similarly).

State populations. Macrostate populations are estimated in two ways. From the trajectory, after dynamical coring (above), as the fraction of frames assigned to each state. From the coarse-grained MSM, using the soft memberships χ_{ik} as weights on the microstate stationary distribution π , the probability of macrostate S_k is computed as:

$$P(S_k) = \sum_i \pi_i \chi_{ik} \quad (1)$$

The membership χ_{ik} down-weights boundary microstates that cannot be confidently assigned to a single set.

State lifetimes. Macrostate lifetimes are estimated from the cored trajectory by collecting the residence times in each state and calculating their median and mean.

III. RESULTS

A. The four pipelines yield valid Markov models that already differ in their projected landscapes and timescales

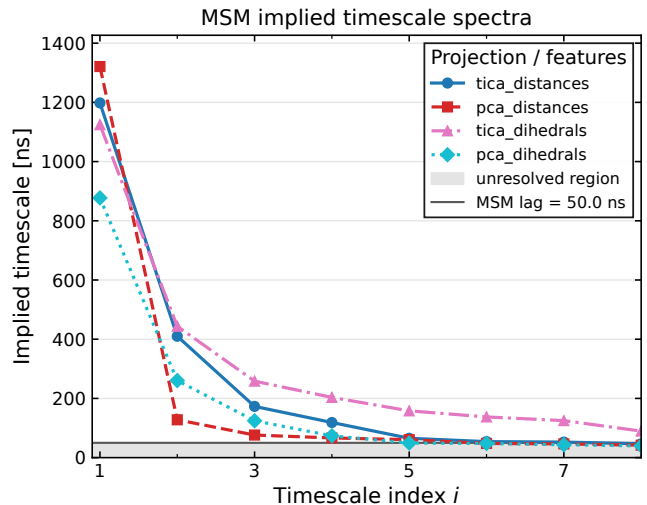


FIG. 3: **Relaxation timescales spectra** at selected lag time $\tau_{\text{MSM}} = 50$ ns. The horizontal grey line marks the MSM resolution limit ($t = \tau_{\text{MSM}}$).

The free energy surfaces obtained by projecting the trajectory onto the first two reduced coordinates suggests that the four pipelines may describe the landscape differently (Fig. 2): the distance-based models yield a single dominant minimum, while both dihedral-based models resolve two, and the tICA projections show an additional

well-separated basin. Very importantly, free energy surfaces do not generally have a simple interpretation in terms of dynamics (e.g. free energy maxima may not be kinetic barriers), therefore these surfaces are only indicative, but the differences anticipate a pipeline dependence in the metastable structure resolved later (Sec. III D, III E).

At the selected common discretization ($n_{\text{clusters}} = 50$, $\tau_{\text{MSM}} = 50$ ns; see Methods), the implied-timescale test (Supp. Fig. SF2) shows that the slowest timescales are approximately converged for all four pipelines. Two observations: first, the implied timescales depend far more strongly on the pipeline (features + reduction method) than on the number of clusters. Second, the full relaxation spectra at $\tau_{\text{MSM}} = 50$ ns (Fig. 3) shows that the PCA pipelines yield systematically lower timescales than the tICA ones, indicating that they approximate the underlying slow dynamics less faithfully, according to the variational principle for Markov processes [15]. This result is reasonable since PCA does not optimize for kinetic separation.

Taken together, these checks show that all four MSMs are approximately Markovian at $\tau_{\text{MSM}} = 50$ ns, $n_{\text{clusters}} = 50$. At the same time, their timescale spectra indicate that the pipelines disagree on how many slow processes exist beyond the dominant one.

B. The dominant slow process relates to the folding–unfolding transition

The dominant implied timescale is reasonably close across most pipelines (Fig. 3). A preliminary inspection of the sign pattern of the first non-trivial right ¹ eigenvector, which closely follows the fraction of native contacts Q (Supp. Fig. SF5), suggests that the slowest resolved mode separates folded from unfolded conformations.

The spectra differ, however, in what lies beyond that dominant process (Fig. 3). The *pca-distances* model shows the largest gap between the first and second timescales, consistent with a strongly metastable, essentially two-state process. The other three show a less pronounced gap, and the *tica-dihedrals* model departs most from a two-state picture: its spectrum decays slowly, with several timescales remaining above the resolution limit ($t = \tau_{\text{MSM}}$) even though its slowest timescale matches the distance-based models.

To compare the models at the coarsest level, I impose a two-state PCCA++ decomposition on each and measure, for every model pair, the fraction of frames the two

models place on opposite sides of the partition (the off-diagonal quadrants of Fig. SF6). At this level the four pipelines are *broadly* consistent: the two distance-based models disagree on only $\simeq 1.6\%$ of frames, and the distance/dihedral PCA pair on $\simeq 2.3\%$. The *tica-dihedral* model exhibits the most difference, disagreeing with each of the others on $\simeq 8\%$ of frames.

For *tica-dihedral*, the origin of this disagreement is visible in the single-model S_1 membership distributions (Supp. Fig. SF6, top): sharply bimodal and near $\{0, 1\}$ for the PCA-based representations, but broad and much fuzzier for *tica-dihedral*. This fuzziness means the model does not resolve a clean boundary between the two basins (unsurprising, given that its spectrum lacks a clear eigenvalue gap). But the boundary sharpness is not the only issue: even the most agreeing model pairs retain a non negligible amount of probability mass in the off-diagonal corners (Supp. Fig. SF6, top)—frames that *both* models assign confidently, yet to opposite sides of the partition. This residual mass indicates that the four pipelines do not resolve exactly the same partition, beyond mere differences in boundary sharpness.

In summary, the pipelines agree on the coarsest description but not on its precise boundary.

C. The folded basin is mostly defined by the hydrophobic core

The populations of S_0 and S_1 (Supp. Fig. SF6) closely match the unfolded and folded populations reported by Piana *et al.* [10], which is consistent with S_1 being the folded basin. To characterize it structurally while avoiding contamination from boundary frames, I define *consensus* microstates as those assigned to S_1 with high confidence ($\chi_{i,1} > 0.8$) by all four pipelines, and restrict the analysis to this subset ($\simeq 61\%$ of the trajectory).

A simple characterization follows from comparing, contact by contact, the distance distributions inside and outside the consensus set. For each contact present in the crystal structure (`hp35.crystaldists`), I score how strongly it distinguishes the two sets by the standardized median difference

$$\text{score}_{i,j} = \frac{\text{median}(r_{ij}^{\text{other}}) - \text{median}(r_{ij}^{\text{consensus}})}{[IQR_{ij}^{\text{other}} + IQR_{ij}^{\text{consensus}}]/2}. \quad (2)$$

where r_{ij} is the distance between residues i and j and IQR stands for interquartile range. A large positive score indicates a contact that is markedly shorter in the consensus set, relative to the spread of its distribution, and therefore characteristic of the folded basin.

The highest-scoring contacts (Fig. 4, bottom) are the long-range pair (10, 28)–(10, 29), which packs helix 1 against helix 3, and the contacts among Phe6, Phe10 and Phe17 that form the hydrophobic core (Fig. 1) and residues close by. Intra-helix contacts, by contrast, do not rank among the discriminating features. This indicates

¹ The right eigenvector must be used since Deeptime represents transition matrices in the row-stochastic convention: $\sum_j T_{ij} = 1$. Under detailed balance, the left eigenvector contains the same information but weighted by the stationary population: $l_i = \text{diag}(\pi) \cdot r_i$.

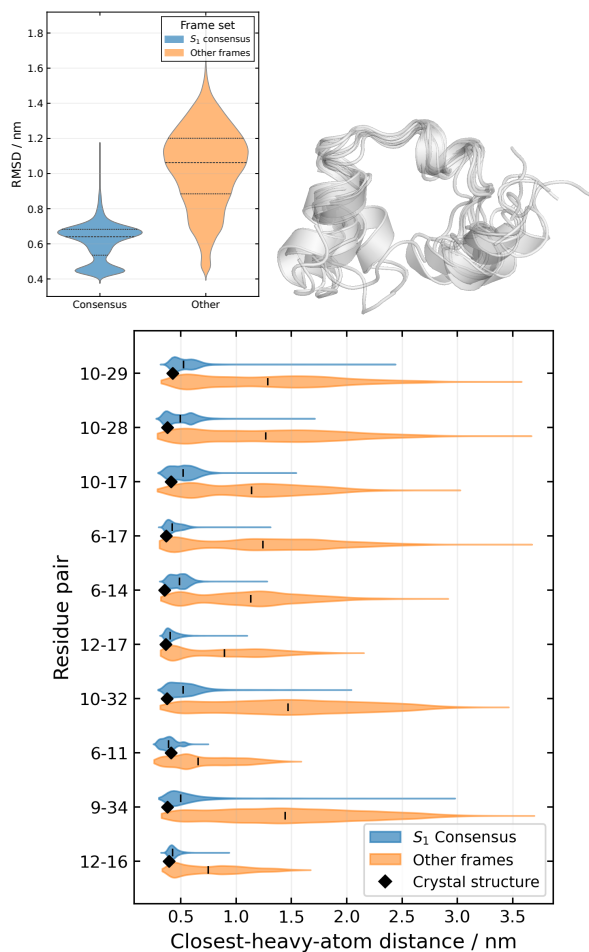


FIG. 4: Structural analysis of S_1 consensus frames. **Top left:** distribution of RMSD from the crystal structure (PDB 2F4K) for consensus frames (blue) and all other frames (orange); dashed lines mark the median and quartiles. **Top right:** overlay of eight randomly chosen consensus frames made with PyMOL [18]. **Bottom:** closest-heavy-atom distance distributions for the ten contacts that best discriminate the two frame sets, ranked by the score of Eq. 2 (most discriminating at the top). For each pair, the vertical tick marks the median of each distribution, and the black diamond marks the distance in the crystal structure.

that appreciable helical structure persists in the unfolded ensemble, whereas hydrophobic-core formation is largely confined to the folded basin. The interpretation is corroborated by comparing all intra-helical contacts inside and outside the consensus set (Supp. Fig. SF7): outside, their distributions are much broader with systematically larger medians, yet still compatible with helical structure, especially for helix 3.

Interestingly, within the consensus ensemble both the RMSD and the (10,28)–(10,29) contacts are bimodal (Fig. 4), which suggest the possibility that the folded basin contains two alternative conformations. This is investigated in Sec. III E.

Taken together, these results indicate that the folded and unfolded ensembles are distinguished primarily by the hydrophobic core and the helix-1–helix-3 packing that encloses it, rather than by helix formation *per se*.

D. Three-states PCCA++ on tICA models reveal a compact non-native state and a loop rearrangement

The analysis so far has treated the system as two-state. The implied-timescale spectra of the tICA models (Fig. 3), however, show additional slow processes that remain above the resolution limit, motivating a finer decomposition. Unlike the *pca-distances* spectrum, the tICA-models spectra display no clean eigenvalue gap, so the number of macrostates cannot really be fixed by the spectral-gap criterion (Methods). In theory, a partition with $n + 1$ macrostates appears legitimate if n is the number of timescales above the resolution threshold. In practice, however, increasing n also tends to produce fuzzier membership probabilities, indicating microstates not confidently assigned to a single macrostate.

In the following, for time constraints reasons I will restrict the discussion to $n = 3$, for the *tica-distances* and the *tica-dihedrals* models, leaving a more detailed analysis and comparison as a possibility for future work.

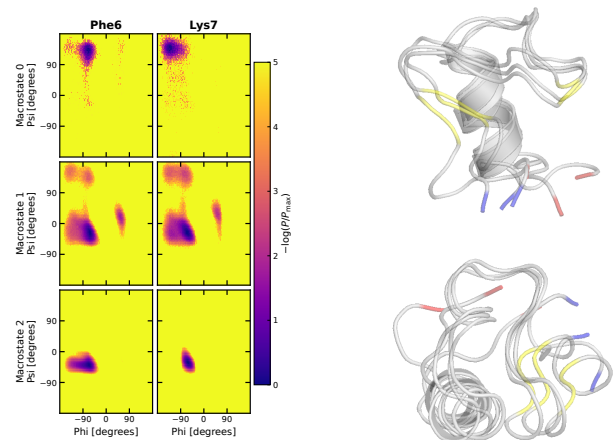


FIG. 5: Structural characterization of macrostate S_0 in the three-state tICA-distance model. Ramachandran distributions of residues Phe6 and Lys7 together with representative conformations from macrostate S_0 . This state is characterized by the formation of helix 3, while the N-terminal region adopts a β -hairpin-like structure in place of helix 1. Residues highlighted in yellow correspond to those shown in the Ramachandran plots. The N-terminus is highlighted in blue, the C-terminus in red. Visualization with PyMOL [18].

tICA-distances. The three-state decomposition adds a sparsely populated ($\sim 1\%$) macrostate, S_0 . It is distinguished from S_1 and S_2 by the backbone dihedrals of

Phe6, Lys7 and Gly15: these residues occupy the β region of the Ramachandran plot in S_0 , sample both the α_R and β regions in S_1 , and are confined to α_R in S_2 (Fig. 5). Structurally, S_0 appears well defined and compact: helix 3 stays folded throughout, while residues 6–14 form a stable β -hairpin-like formation in place of helix 1. This non-native topology makes S_0 a plausible off-pathway intermediate, i.e. a putative kinetic trap. Consistent with a rare but long-lived state, the cored macrostate trajectory shows a single visit, lasting approximately $1 \mu\text{s}$.

tICA-dihedrals. Here the three-state decomposition splits the space into S_0 ($\simeq 6\%$), S_1 ($\simeq 25.8\%$) and S_2 ($\simeq 68.1\%$). S_1 and S_2 recover the unfolded and folded/partially folded basins of the two-state analysis (there labelled S_0 and S_1 , respectively), and S_0 is the new state. Its structural signature is less obvious than in the distance model. Comparing the dihedral distributions (Supp. Fig. SF9), the clearest difference between S_0 and the other two states is confined to the Leu20–Pro21 loop joining helices 2 and 3 in the native fold: in S_0 the ϕ/ψ angles of Leu20 lie predominantly in the β region, whereas in S_1 and S_2 they concentrate in α_R . From the cored trajectory, S_0 has an empirical population near 3% and a residence time of $0.32 \mu\text{s}$ (median; $0.52 \mu\text{s}$ mean). It is entered exclusively from S_1 and returns to S_1 ; no $S_1 \rightarrow S_0 \rightarrow S_2$ passages are observed. Overall, it appears that the $S_0 \leftrightarrow S_1$ transition corresponds to a local rearrangement of the second loop within the unfolded ensemble.

E. Dihedral-based models resolve substructure within the folded basin

The previous analysis (Section III C) showed that several observables, including the RMSD and specific long-range contacts such as 10–29, display bimodal distributions within the consensus folded ensemble. The same bimodality is visible in the free-energy projections obtained from the dihedral-based models (Fig. 2) and is reflected in the distribution of the PCCA++ membership probability for macrostate S_1 in the two-state decomposition (Supp. Fig. SF6): These observations suggest that the folded basin identified by the dihedral-based MSMs is itself heterogeneous and may contain two structurally distinct substates. However PCCA++, at least with 3 states, did not identify finer structure within the folded basin.

To investigate this further, I performed only a simple and exploratory analysis, which nevertheless yielded interesting results. Specifically, I moved beyond the PCCA++ macrostate assignment and heuristically subdivided the folded ensemble identified by the dihedral-based models according to the distinct peaks observed in the distribution of PCCA++ membership probabilities.

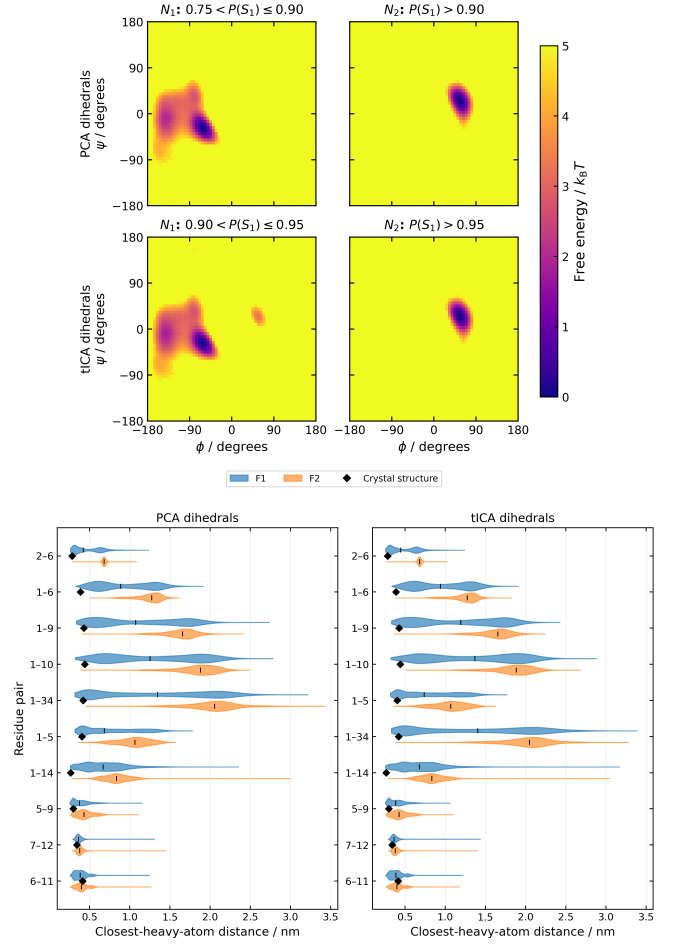


FIG. 6: **Structural distinction between the two folded substates identified by the dihedral-based MSMs.** *Top:* (ϕ, ψ) distribution of the residue Asp3 in the two subpopulations corresponding to the distinct minima observed within the folded basin. *Bottom:* closest-heavy-atom distance distributions for the ten contacts that best discriminate F_1 and F_2 , ranked by the absolute value of the score Eq. 2 (most discriminating at the top). For each pair, the vertical tick marks the median of each distribution, and the black diamond marks the distance in the crystal structure.

(see Supp. Figure SF6):

$$\begin{cases} \text{pca-dihedrals:} & F_1 : \{x_i : 0.75 < \chi_{i,1} < 0.9\}, \\ & F_2 : \{x_i : 0.9 < \chi_{i,1}\}, \\ \text{tica-dihedrals:} & F_1 : \{x_i : 0.9 < \chi_{i,1} < 0.95\}, \\ & F_2 : \{x_i : 0.95 < \chi_{i,1}\}. \end{cases}$$

The two subsets of microstates (F_1, F_2) are found to be distinguished primarily by the conformational state of a single residue, Asp3. In one state, Asp3 predominantly occupies the α_R region of Ramachandran space, whereas in the other it preferentially populates the α_L region (Fig. 6, top). Furthermore, an analysis of contact importance with the same score used before (Eq. 2) indi-

cates that the most important contacts are those involving the N terminus *Leu1* (6, bottom), and suggest the interpretation of these two substates as a "locked" and "unlocked" state (F_1 and F_2 respectively). Both substates are individually metastable: from trajectory analysis (Supp. Fig. SF10, top), their residence-time distributions are close to single-exponential (Supp. Fig. SF10, bottom), with a median residence time of $\simeq 0.4 \mu\text{s}$ (mean $\simeq 0.6 \mu\text{s}$; Supp. Table I), comparable to that of the loop macrostate that three-state PCCA++ does resolve on *tica-dihedral* ($0.32 \mu\text{s}$; Sec. IIID).

The fact that PCCA++ resolves the loop-rearrangement state but not these substates is not necessarily contradictory. The latter may correspond to finer kinetic subdivisions that would probably emerge in a PCCA++ decomposition with more than three macrostates.

IV. DISCUSSION AND CONCLUSIONS

This project set out to characterize the conformational ensemble of the HP35 Nle/Nle mutant from a structural standpoint. To limit the bias inherent in any single MSM pipeline, four models were built and compared, varying the feature set and the dimensionality-reduction step. Beyond the specific findings for HP35, the analysis illustrates how different MSM pipelines can lead to different interpretations of the same simulation data, even for a relatively small protein and a high-quality trajectory.

Taken together, the four models yield the following picture of the HP35 folding landscape:

1. A dominant slow process separates two metastable basins: an unfolded basin ($\simeq 31\text{--}32\%$) and a folded/partially folded basin ($\simeq 67\text{--}68\%$). The two are distinguished not by secondary-structure content but mainly by the presence or absence of the native hydrophobic-core contacts and the close packing of helices 1 and 3 (Sec. IIIC);
2. A low-population ($\simeq 1\%$), compact metastable state with a lifetime of $\sim 1 \mu\text{s}$, in which helix 3 and the helix-2–helix-3 loop are folded but an extended, hairpin-like segment replaces helix 1 (Sec. IIID);
3. Two substates within the folded basin, F_1 and F_2 , of comparable population ($33.8\text{--}35\%$ and $31\text{--}32\%$, respectively). They differ in the configuration of the N-terminus and are discriminated mainly by the (ϕ, ψ) angles of Asp3 (α_R in F_1 , α_L in F_2) and by the presence (F_1) or absence (F_2) of several native contacts involving *Leu1* (Sec. IIIE).

This picture agrees on several points with the macrostate model of Jain and Stock [4], obtained from

a MSM construction pipeline which used dihedral representation, PCA, and coarse graining with most-probable-path clustering (MPP). Their intermediate I_1 corresponds closely to the substate F_2 found here. They also report an off-pathway, partially folded state I_3 in which an extended conformation replaces helix 1; this matches the putative kinetic trap resolved by my *tica-distance* model, both structurally and in population (1.49% vs. $\simeq 1\%$ here). On these points the present analysis supports their description, and it further shows that I_3 is recoverable with a standard tICA/PCCA++ pipeline: here, specifically when the features are contacts rather than dihedrals.

None of the models, at least within the present analysis, did resolve the alternative folded conformation characterized by disrupted contacts in the C-terminal region, which has been observed experimentally [3] and also recovered by the MSM study by Jain and Stock. It remains possible that this conformation would also be recovered within the present workflow by employing a PCCA++ decomposition with a larger number of macrostates.

Although the present analysis provided several insights, it remains far from complete. From a structural perspective, the natural next step would be to investigate PCCA++ decompositions with a larger number of macrostates, particularly for the tICA-based models, and to perform a more systematic comparison of the identified metastable states. A more complete kinetic characterization would in turn require analysis at the microstate level, including the application of transition path theory (TPT) to identify dominant folding pathways and quantify the flux between metastable states.

V. REFERENCES

Data availability statement: Preprocessed trajectories of the HP35 villin headpiece are kindly made available by Nagel et al. and downloadable at their GitHub <https://github.com/moldyn/HP35-DESRES>. All code necessary to replicate this analysis is available on GitHub at <https://github.com/miriamzara/MSM>.

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Appendix A: Supplementary Figures

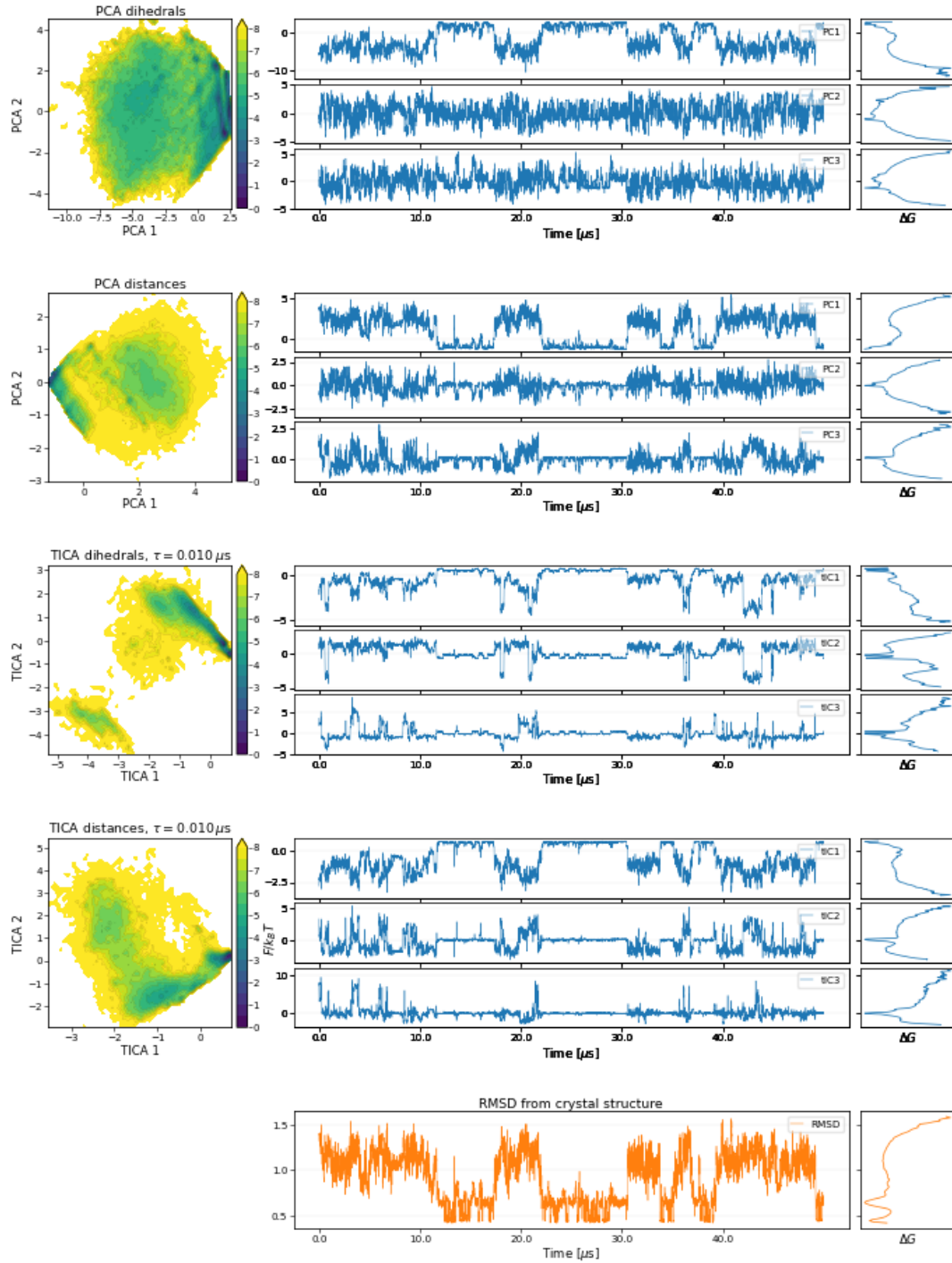


FIG. SF1: **Reduced-coordinate representations obtained from the four feature-dimensionality-reduction pipelines.** Rows correspond to PCA dihedrals, PCA contact distances, tICA dihedrals, and tICA contact distances. For each pipeline, the left panel shows the projection of the trajectory onto the first two reduced coordinates. The remaining panels show the time evolution of the first three reduced coordinates during the first $\sim 40 \mu\text{s}$ of simulation together with the corresponding one-dimensional free-energy profiles computed from the full trajectory. The bottom row reports the RMSD from the crystal structure and its free-energy profile for comparison.

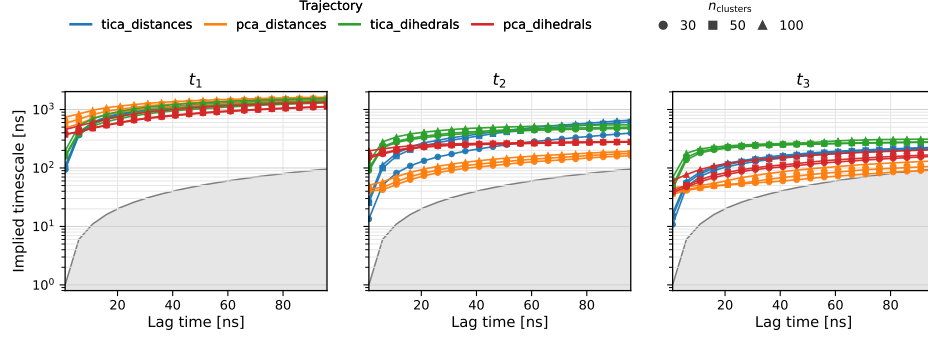


FIG. SF2: **Results of the timescale convergence test:** First three implied timescales as a function of the MSM lagtime. Colors denote the four combinations of feature set and dimensionality reduction method, while marker styles indicate the number of microstates used in the state-space discretization. The shaded region corresponds to $t_i \leq \tau_{MSM}$, where t_i is the implied timescale and τ_{MSM} the MSM lagtime. Timescales falling within this region are not resolved by the model, as the associated relaxation process occurs on a timescale shorter than a single Markov-chain transition step.

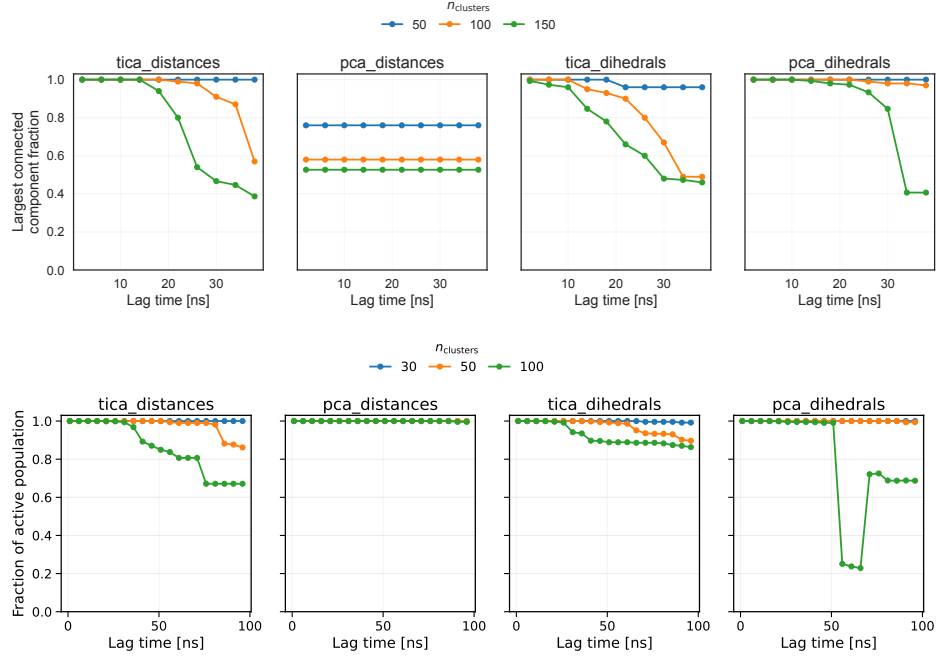


FIG. SF3: **MSM connectivity and active population as a function of lag time.** Top panel: fraction of microstates in the largest connected component of the MSM as a function of lag time. Bottom panel: fraction of the total equilibrium population contained in the largest connected component of the MSM as a function of lag time

Model	State	N_{visits}	Mean [μs]	Median [μs]	Population
tica_dihedrals	F1	177	0.603	0.463	0.350
tica_dihedrals	F2	145	0.658	0.435	0.313
pca_dihedrals	F1	195	0.529	0.412	0.338
pca_dihedrals	F2	145	0.676	0.480	0.321

TABLE I: **Analysis of subsets F_1 and F_2 "manually" identified from dihedral-based models.** Residence times in the folded sub-basins F_1 and F_2 , computed from the cored microstate trajectory of each dihedral model.

N_{visits} is the number of continuous visits to the sub-basin, and the population is the fraction of cored frames assigned to it.

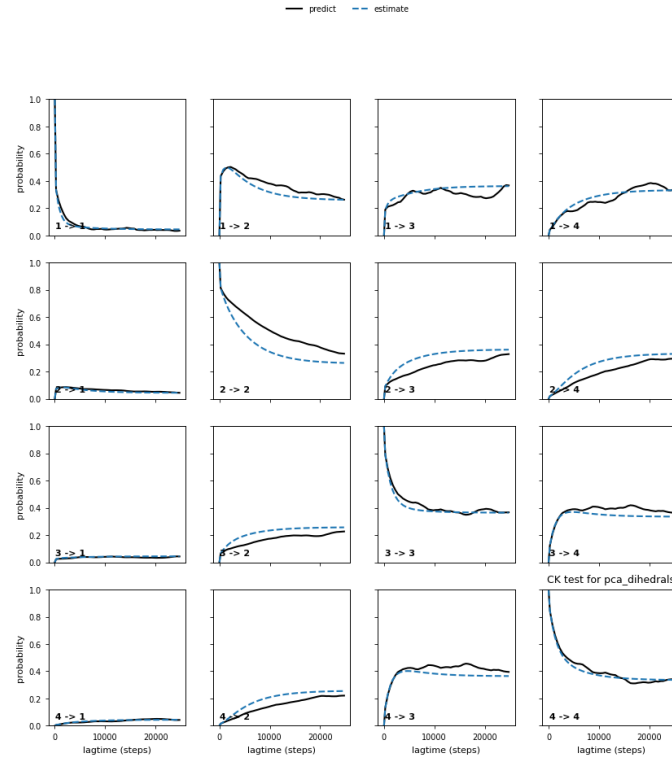


FIG. SF4: **Chapman-Kolmogorov test for the *pca*-dihedrals MSM.** The microstate model estimated at $\tau = 50$ ns was lumped into four metastable sets with PCCA++. Each panel ($i \rightarrow j$) shows the probability of being in set j at time $k\tau$ given the system started in set i . Solid lines: MSMs estimated independently at each lag; dashed lines: predictions from the $\tau = 50$ ns model. Close agreement indicates approximate Markovianity at the chosen lag.

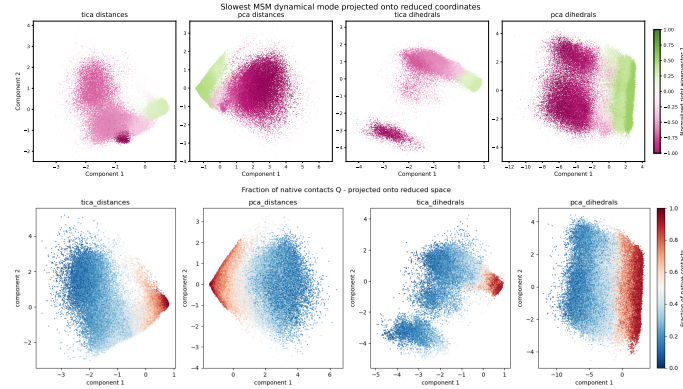


FIG. SF5: **Comparison between the slowest MSM relaxation mode and a structural folding descriptor.** Top row: values of the first non-trivial *right*(*) eigenvector projected onto the first two components of the reduced coordinate space. Positive and negative components identify regions that exchange probability on the slowest resolved timescale of the Markov model. Bottom row: projection of the fraction of native contacts, Q , onto the same coordinates. The close correspondence between the eigenvector sign pattern and the distribution of Q indicates that the slowest kinetic process captured by the MSM is associated with the folding-unfolding transition.

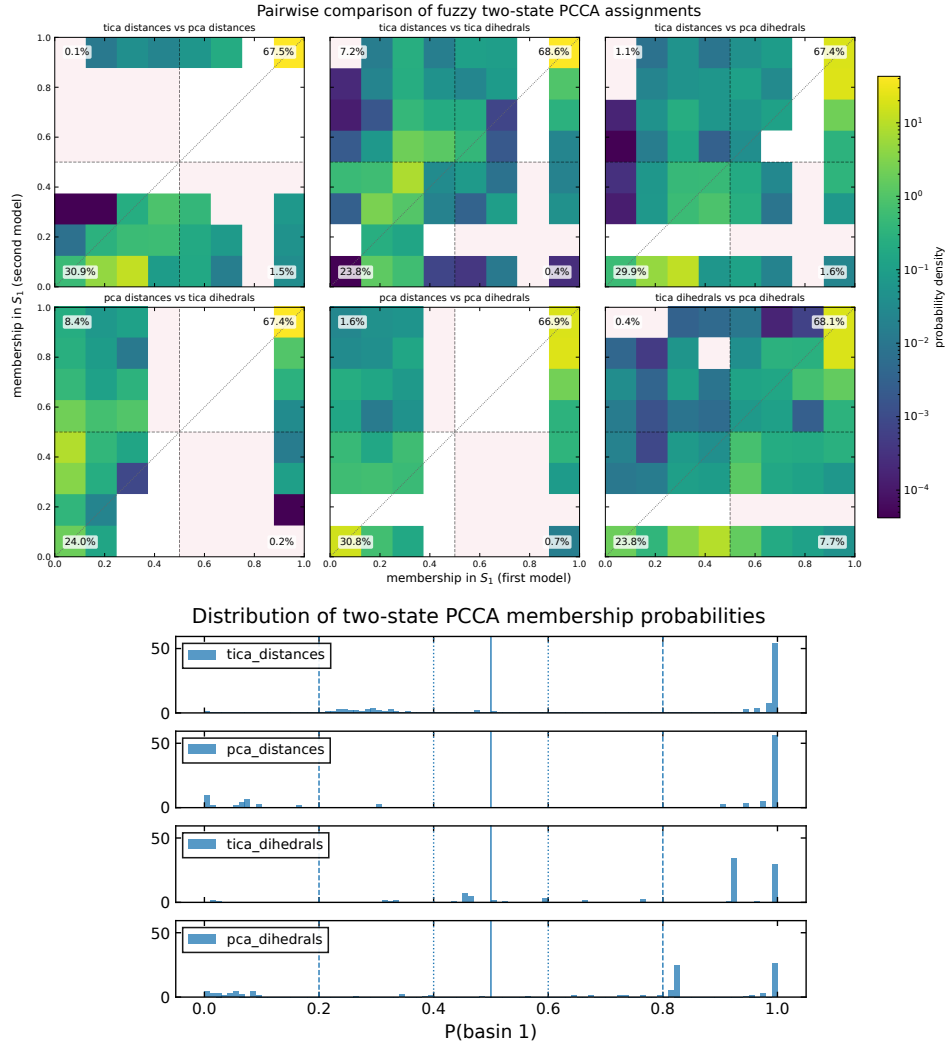


FIG. SF6: **Fuzzy membership probabilities of the trajectory frames to the two macrostates across models.** Top: Each panel shows the fuzzy membership probability of each frame in the trajectory to macrostate S_0 ($\simeq$ denatured basin) and macrostate S_1 ($\simeq$ folded basin) for a different model. The models based on distances (tica-distances and pca-distances) show very similar distributions, while the tica-dihedrals model exhibits a broader distribution, particularly for frames assigned to the denatured basin. This indicates that the tica-dihedrals model has more uncertainty in assigning frames to the denatured basin. Bottom: Distribution of the fuzzy membership probability of each frame to macrostate 1 for the four MSMs. Probabilities close to 0 or 1 correspond to frames that are confidently assigned to a single macrostate.

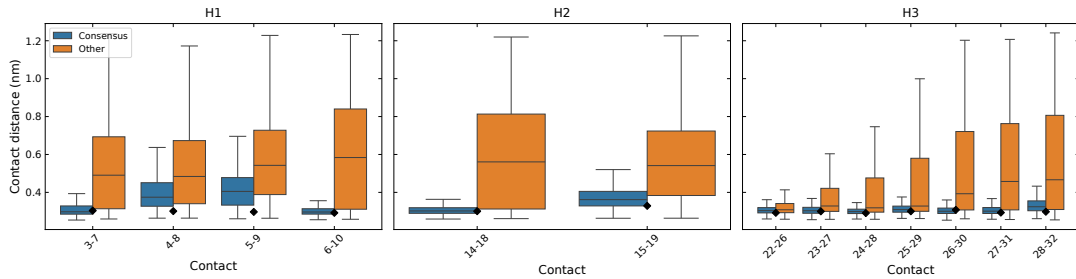


FIG. SF7: **Helix content in the consensus S_1 frames.** Distribution of distances for all intra-helical contacts in the consensus folded ensemble (S_1 , blue) and in all other frames (orange). The black squares marks the distance observed in the crystal structure PDB 2F4K. Although helical contacts are generally more compact within S_1 , their distributions overlap substantially with those of the remaining frames, indicating that significant helical structure is present both inside and outside the folded basin.

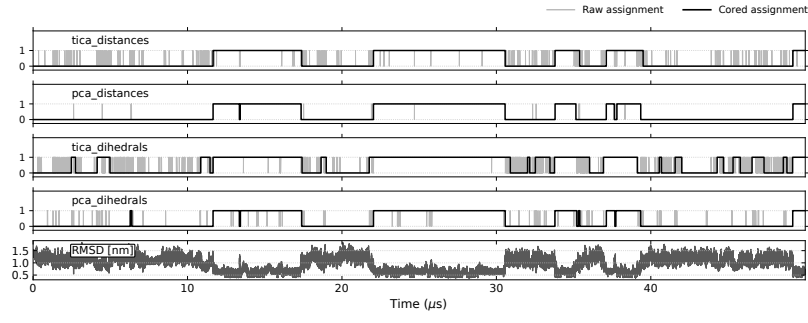


FIG. SF8: **Comparison of raw and cored macrostate trajectories for the two-state PCCA++ models.** For each MSM, the grey trace shows the instantaneous hard assignment of each frame, while the black trace shows the filtered assignment obtained by removing transitions that persist for less than one Markov lag time ($\tau = 50$ ns). The RMSD time series is shown at the bottom for reference. This filtering suppresses rapid recrossings near the state boundary and highlights the long-lived folded and unfolded episodes underlying the dynamics. Only the first $\simeq 50 \mu s$ of the trajectory are shown.

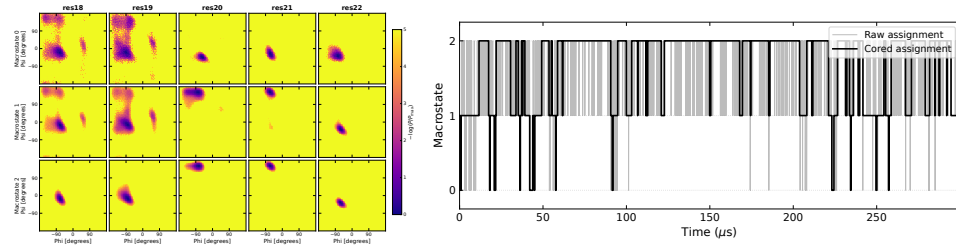


FIG. SF9: **Structural characterization of the 3-state PCCA macrostates for the tica-dihedrals model.** Top: (ϕ, ψ) distribution of the residues 18 – 22 in the three macrostates S_0, S_1, S_2 . Bottom: trajectory of the three-state PCCA++ model for tICA-dihedrals, with hard assignments and cored assignments (transitions are counted if they remain in the destination for at least one markov step, which is $\tau = 50$ ns).

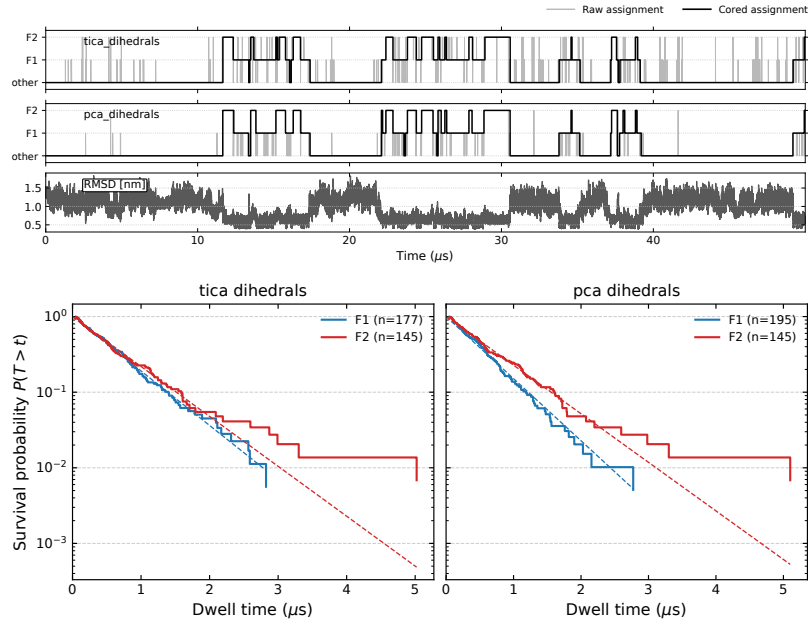


FIG. SF10: **Analysis of subsets F_1 and F_2 "manually" identified from dihedral-based models.** Top: first $\simeq 40 \mu s$ of the discretized trajectory (F_1, F_2, other) highlight $F_1 \leftrightarrow F_2$ transitions. Bottom: survival probability (1-cdf) and linear fit shows the distribution of the residence time is well approximated by an exponential.

Appendix B: Supplementary Methods

S1. Derivation of PCA and tICA

The presentation of tICA in this section follows closely that of [17].

PCA and tICA are both linear dimensionality reduction techniques, i.e. they project the data onto a linear subspace of the original feature space. Let $\{\mathbf{x}(t)\}_{t=0}^{T-1}$ be a multidimensional, discrete time-series, where each snapshot is a column vector, of dimension D , corresponding to an arbitrary, vectorized representation of a protein conformation (e.g. a set of dihedral angles). Suppose the data has been sampled from a *stationary* and *time-reversible* stochastic process. Without loss of generality, I assume that the data have been centered, $\mathbb{E}[\mathbf{x}] = 0$. Using the bracket notation $\langle x | := \mathbf{x}^T$, $|x\rangle := \mathbf{x}$, I define the covariance matrix:

$$\mathbf{C}_0 := \mathbb{E}[|x\rangle \langle x|] \quad (\text{B1})$$

and the time-lagged covariance matrix:

$$\mathbf{C}_{0,\tau} := \mathbb{E}[|x(t)\rangle \langle x(t+\tau)|]. \quad (\text{B2})$$

For a stationary reversible process,

$$\mathbf{C}_{0,\tau} = \mathbf{C}_{\tau,0}^T = \mathbf{C}_{\tau,0},$$

so that the lagged covariance matrix may be denoted simply by $\mathbf{C}_\tau$.

If $|\hat{u}\rangle \langle \hat{u}|$ is a projection operator (i.e. $|\hat{u}\rangle$ is a unit vector), then:

$$\langle \hat{u} | \mathbf{C}_0 | \hat{u} \rangle = \mathbb{E}[\langle \hat{u} | x \rangle \langle x | \hat{u} \rangle] = \mathbb{E}[\langle \hat{u} | x \rangle^2] \quad (\text{B3})$$

is the variance of the projection of the data along $|\hat{u}\rangle$, and

$$\begin{aligned} \text{ACF}_{\hat{u}}(\tau) &= \frac{\langle \hat{u} | \mathbf{C}_\tau | \hat{u} \rangle}{\langle \hat{u} | \mathbf{C}_0 | \hat{u} \rangle} \\ &= \frac{\mathbb{E}[\langle \hat{u} | x(t) \rangle \langle \hat{u} | x(t+\tau) \rangle]}{\mathbb{E}[\langle \hat{u} | x \rangle^2]} \end{aligned} \quad (\text{B4})$$

its autocorrelation at lag time τ .

PCA seeks the direction of maximal variance. The first principal component is defined as

$$\begin{aligned} |\hat{u}_1\rangle &= \arg \max \langle \hat{u} | \mathbf{C}_0 | \hat{u} \rangle \\ \text{subject to } \|\hat{u}\| &= 1 \end{aligned} \quad (\text{B5})$$

this optimization problem can be solved introducing a Lagrange multiplier λ , and the stationarity condition yields

$$\mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i |\hat{u}_i\rangle, \quad (\text{B6})$$

i.e. the projection vector is an eigenvector of the covariance matrix. The corresponding eigenvalue equals the variance of the data projected onto the principal component:

$$\langle \hat{u}_i | \mathbf{C}_0 | \hat{u}_i \rangle = \lambda_i \cdot \langle \hat{u}_i | \hat{u}_i \rangle = \lambda_i.$$

Then the second principal component is defined as the direction of maximal variance orthogonal to the first, giving the

optimization problem:

$$\begin{aligned} |\hat{u}_2\rangle &= \arg \max \langle u | \mathbf{C}_0 | u \rangle \\ \text{subject to } \langle u | u \rangle &= 1 \\ \text{and } \langle u | \hat{u}_1 \rangle &= 0 \end{aligned} \quad (\text{B7})$$

which again can be solved with Lagrange multipliers. Iterating, this implies that the principal components form an orthonormal basis of eigenvectors of the covariance matrix, i.e. they diagonalize the covariance matrix. Dimensionality reduction ($d < D$) is achieved by taking the components along the first d principal components $\{|\hat{u}_1\rangle, \dots, |\hat{u}_d\rangle\}$.

$$z_i = \langle \hat{u}_i | x \rangle, \quad \mathbf{z} = U^T \mathbf{x}. \quad (\text{B8})$$

This is a linear projection onto an orthonormal basis:

$$|x\rangle \rightarrow |z\rangle := \sum_{i=1}^d \langle x | \hat{u}_i \rangle |\hat{u}_i\rangle$$

By construction, in the reduced space components are uncorrelated and ordered by decreasing variance:

$$\mathbb{E}[|z\rangle \langle z|] = \text{diag}(\lambda_1, \dots, \lambda_d) \quad (\text{B9})$$

tICA Unlike PCA, tICA [17] seeks directions $|u\rangle$ that maximize the autocorrelation Eq.(B4) at a fixed lag time τ . The lagtime τ is a free parameter. The autocorrelation is invariant under scaling of the projection vector $|u\rangle$. Differently from PCA, the normalization constraint is usually implemented as $\langle u | \mathbf{C}_0 | u \rangle = 1$, as this choice simplifies the optimization problem. Formally, tICA solves the optimization problem:

$$\begin{aligned} |\hat{u}_1\rangle &= \arg \max \langle u | \mathbf{C}_\tau | u \rangle \\ \text{subject to } \langle u | \mathbf{C}_0 | u \rangle &= 1 \end{aligned} \quad (\text{B10})$$

using the same procedure as for PCA, I can solve this with Lagrange multipliers, and show that

$$\mathbf{C}_\tau |\hat{u}_1\rangle = \lambda_1 \mathbf{C}_0 |\hat{u}_1\rangle, \quad (\text{B11})$$

The generalized eigenvalue λ_i is equal to the autocorrelation of the projected coordinate $\langle \hat{u}_i | x(t) \rangle$ at lag time τ . Assuming that this coordinate approximates a single relaxation mode of the dynamics, its autocorrelation decays exponentially, yielding

$$\lambda_i \approx e^{-\tau/t_i}, \quad (\text{B12})$$

where t_i is the corresponding relaxation timescale. In practice, the relation is approximate because a tIC may contain contributions from multiple dynamical modes.

Similarly as done with PCA, I can iterate this procedure and define the second tIC as the direction of maximal autocorrelation at lag time τ that is *uncorrelated* with the first:

$$\begin{aligned} |\hat{u}_2\rangle &= \arg \max \langle u | \mathbf{C}_\tau | u \rangle \\ \text{subject to } \langle u | \mathbf{C}_0 | u \rangle &= 1 \\ \text{and } \langle u | \mathbf{C}_0 | \hat{u}_1 \rangle &= 0 \end{aligned} \quad (\text{B13})$$

the solution $|\hat{u}_1\rangle$ is again a solution of the same generalized eigenvalue problem as in Eq.(B10). generalized eigenvalue

problem. Dimensionality reduction ($d < D$) is achieved by taking the components of the data $\mathbf{x}(t)$ along the first d tICs $\{|\hat{u}_1\rangle, \dots, |\hat{u}_d\rangle\}$.

$$\mathbf{z}(t) = (\langle x(t)|\hat{u}_1\rangle, \dots, \langle x(t)|\hat{u}_d\rangle) \quad (\text{B14})$$

By construction, in the reduced space components z_i have unit variance, are uncorrelated at zero lag, and are ordered by decreasing autocorrelation at lag time τ .

$$\mathbb{E}[|z\rangle \langle z|] = \mathbb{I} \quad (\text{B15})$$

tICA as a diagonalization in the whitened coordinate space It is worth noting that unlike in PCA, the tIC vectors are generally not orthogonal under the standard Euclidean inner product. Instead, as I said, they satisfy the generalized orthogonality relation

$$\langle u_i | \mathbf{C}_0 | u_j \rangle = \delta_{ij}.$$

i.e. they corresponding components z_i and z_j are *uncorrelated* at zero lag. Nevertheless, the tICA procedure can be reformulated as a standard eigenvalue problem in a transformed coordinate system. The key idea is to first apply a whitening transformation, which rescales and rotates the original data so that all directions have unit variance and are mutually uncorrelated. Define the ZCA whitening transformation

$$|y\rangle = \mathbf{C}_0^{-1/2} |x\rangle$$

where $\mathbf{C}_0^{-1/2}$ is symmetric and positive definite. By construction, the covariance matrix in the transformed space is the identity:

$$\mathbb{E}[|y\rangle \langle y|] = \mathbf{C}_0^{-1/2} \mathbf{C}_0 \mathbf{C}_0^{-1/2} = \mathbf{I}.$$

and time-lagged covariance matrix in the whitened space is

$$\hat{\mathbf{C}}_\tau = \mathbf{C}_0^{-1/2} \mathbf{C}_\tau \mathbf{C}_0^{-1/2}$$

$\hat{\mathbf{C}}_\tau$ is symmetric and can thus be diagonalized:,

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle, \quad \langle v_i | v_j \rangle = \delta_{ij}.$$

If $|u_i\rangle$ solves the tICA problem (B10), then $|v_i\rangle = \mathbf{C}_0^{1/2} |u_i\rangle$ is an eigenvector of $\hat{\mathbf{C}}_\tau$ with the same eigenvalue.

Indeed, multiplying the generalized eigenvalue problem (B10) from the left by $\mathbf{C}_0^{-1/2}$ and defining $|v_i\rangle = \mathbf{C}_0^{1/2} |u_i\rangle$ yields

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle.$$

Consequently, the tICA components z_i correspond to the orthogonal projection of the whitened data $\mathbf{y}$ onto the orthonormal basis defined by $\{|v_i\rangle\}$:

$$z_i(t) = \langle x(t) | u_i \rangle = \langle x(t) | \mathbf{C}_0^{-1/2} v_i \rangle = \langle y(t) | v_i \rangle.$$

Alternative generalized eigenvector normalizations for tICA

The construction presented above corresponds to the original formulation of tICA, in which the covariance matrix of the transformed data is the identity (B15). However, alternative rescaling of the transformed coordinates have been used in the literature and are usually implemented in softwares to endow Euclidean distances in the reduced space with a more

direct kinetic interpretation. These can be particularly useful when the tICA coordinates are subsequently clustered to construct a Markov state model, as I do in this work.

The most commonly used choice is the *kinetic map* in which each tIC component z_i is multiplied by its corresponding eigenvalue,

$$\tilde{z}_i(t) = \lambda_i z_i(t). \quad (\text{B16})$$

With this normalization, the coordinates representing the slowest processes "weight more", and a unit step in those directions in the reduced space is assigned a larger distance. An alternative is the *commute map*, where each coordinate is rescaled according to its implied timescale,

$$\tilde{z}_i(t) = \sqrt{\frac{t_i}{2}} z_i(t), \quad t_i = -\frac{\tau}{\ln \lambda_i}. \quad (\text{B17})$$

With this normalization, Euclidean distances approximate commute distances, i.e. the expected time required to travel from one configuration to another and back. This is a variation on the same idea behind the kinetic map, being eigenvalues and timescales connected by (B12). Both normalizations are defined in Deeptime [11].